MICOANDAR JUSTINUS MARBUN

SOFTWARE ENGINEER

Web, Mobile & AI Application Development

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PROFESSIONAL SUMMARY

Fresh Graduate with a Bachelor's Degree in Informatics from Universitas Gunadarma (2026), focused on Software Engineering. Experienced through academic and independent projects in building end-to-end web applications, from interfaces, backend and REST APIs, and database design to application security and deployment, as well as integrating AI models (YOLO) into web and Android applications. Proficient in PHP, JavaScript, React, Python, and MySQL for building functional and responsive software. Highly motivated to learn and ready to contribute to developing functional, secure, and user-friendly software solutions.

SELECTED PROJECTS

Restaurant Ordering System "Restoran Kita"
PHP Native | MySQL | Bootstrap 5 | JavaScript | AJAX/Fetch API | Chart.js | Flatpickr

- Built an end-to-end web food ordering system: session-based cart, checkout, real-time order status tracking (AJAX polling), and an interactive admin dashboard (Chart.js, menu CRUD, Excel export, invoice printing).
- Developed a rule-based chatbot without external APIs, featuring database-driven FAQs, dynamic queries to menu/order tables, and 14 quick replies.
- Implemented security measures: bcrypt with automatic MD5 migration, prepared statements, CSRF protection, and session regeneration.

AI Workplace Safety Monitoring System
React 19 | Vite | Recharts | FastAPI | YOLO11 | PyTorch | OpenCV | MySQL | SQLAlchemy | Docker | Vercel | Railway

- Trained YOLO11s for PPE compliance detection (5 classes, 6,827 images), achieving 91.35% mAP@50, 92.5% precision, and 86.3% recall.
- Built a modular FastAPI backend (rate limiting, security headers, MySQL via SQLAlchemy) and a React frontend with a trend dashboard, upload/live webcam detection (2.5 FPS), and detection history.
- Deployed end-to-end: frontend on Vercel, Dockerized backend and MySQL on Railway.

AI CCTV Person Detection & Monitoring System
React.js | Vite | Tailwind CSS | Recharts | FastAPI | YOLO26n | PyTorch | OpenCV | WebSocket | MySQL

- Trained a custom YOLO26n for person detection on 7,096 CCTV images (50 epochs, Colab T4), achieving 94.5% mAP50, 93.9% precision, and 87.1% recall.
- Built a monitoring application with real-time webcam input (WebSocket), image upload, and video upload, equipped with a statistics dashboard, filtered history, and trend analytics.
- Integrated a context-aware chatbot powered by queries against live data in MySQL.

Food Ingredient Detection & Diet Recipe Recommendation App - Thesis
Kotlin | Android | YOLOv8n (TFLite) | Firebase Firestore | Gemini API

- Developed an Android application with real-time food ingredient detection (YOLOv8n + TensorFlow Lite) and rule-based recipe matching from Firebase Firestore.
- Integrated the Gemini API for cooking instructions and ingredient substitution suggestions.
- Model trained on 3,793 images, achieving 0.901 precision, 0.901 recall, and 0.944 mAP@50.

Student Grade Processing System - SMK Harapan Bangsa
PHP Native | MySQL | HTML5 | CSS3 | JavaScript

- Built an academic information system with three roles (Admin, Teacher, Student): data CRUD, grade entry, automatic final grade calculation (30% assignments, 30% midterm, 40% final exam), and graduation status.
- Provided reports with PDF/Excel export, and implemented session authentication, RBAC, and password hashing.

EDUCATION

Bachelor's Degree - Universitas Gunadarma, Bekasi, West Java, Indonesia 2022 - 2026
Informatics - GPA 3.76 / 4.00
Thesis Title: Application of YOLOv8 for Food Ingredient Detection in an Android-Based Diet Recipe Recommendation System
Scientific Paper Title: Automated Trash Bin System Using Ultrasonic Sensors Based on the Internet of Things

Vocational High School - SMK Telekomunikasi Telesandi Bekasi, West Java 2019 - 2022
Computer and Network Engineering - Final Score 85.51
Industrial Work Practice (Internship) at PPTIK ITB. Report: DHCP Server Installation and Configuration on Ubuntu 16.04 Using VirtualBox (Score 87)

SKILLS

Programming Languages: Python, PHP, JavaScript, TypeScript, Java, Go, SQL.
Frontend: React, Next.js, Vite, Bootstrap 5, Tailwind CSS, Chart.js, Recharts, AJAX / Fetch API.
Backend & API: Node.js, Express.js, FastAPI, Laravel, SQLAlchemy, REST API.
AI & Computer Vision: Computer Vision, YOLO (v8/v11/26), PyTorch, OpenCV, TensorFlow Lite, Gemini API.
Mobile Development: Kotlin, Flutter, Android Studio.
Database: MySQL, PostgreSQL, MongoDB, Oracle Database, Firebase Firestore.
DevOps & Cloud: Docker, Git / GitHub, Railway, Vercel, Postman.

CERTIFICATIONS

Programmer - National Professional Certification Agency (BNSP) / LSP	2026
Skills Competency Test - Computer and Network Engineering	2022
Certificate of Expertise Level II - Computer and Network Engineering Competency	2021

TRAINING

Programmer Course - Universitas Gunadarma	2025
Intermediate Golang - Universitas Gunadarma	2025
Intermediate Oracle - Universitas Gunadarma	2025
Application Development Design - Universitas Gunadarma	2024
Wireless LAN Installation - Universitas Gunadarma	2024